UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 21826

LH LH THE LIFE AND DEATH OF STARS

SEMESTER 1 EXAMINATIONS 2021/22

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. (a) Describe the key qualitative difference between the proton-proton chains and the CNO cycles. What are their approximate power-law dependences on temperature? For roughly what mass ranges does each reaction chain dominate in main-sequence stars? **[6]**

(b) The reaction rate r for a non-resonant nuclear reaction can be approximated by

$$r = r_0 (kT)^{-2/3} \exp[-3(E_G/4kT)^{1/3}],$$

where r_0 is a constant, T is the temperature and E_G is the Gamow energy. Derive an expression for the logarithmic rate of change of r with temperature (i.e., $d \ln r / d \ln T$). Hence, using the table below, determine the temperature power law indices of the pp chains and the CNO cycle at temperature $T = 2 \times 10^7\,\mathrm{K}$.

Why would this approximation for the reaction rate not apply to the triple alpha process? **[12]**

Reaction	E_G
$p(p, e^+\nu)d$	$0.490\,\mathrm{MeV}$
$^{14}\mathrm{N}(p,\gamma)^{15}\mathrm{O}$	$44.8\,\mathrm{MeV}$

(c) Given that the radiative temperature gradient is $\nabla_{\mathrm{rad}} = (d \ln T / d \ln P)_{\mathrm{rad}}$, what do these two power law relations imply about the convective stability of main-sequence cores in stars of $1\,\mathrm{M}_\odot$ or $2\,\mathrm{M}_\odot$? What observable consequences does this have for the tracks these stars follow in the Hertzsprung–Russell diagram immediately after leaving the main sequence? **[5]**

(d) The very first generation of stars must have formed with *no* metals (i.e., metal abundance $Z = 0$) and a hydrogen abundance of $X \approx 0.76$. In light of the different power laws derived above, how might the main-sequence core of a zero-metallicity star of mass $1.5\,\mathrm{M}_\odot$ differ from that of a "modern" star with metallicity $Z \approx 0.02$? How would this star and its evolution differ overall from it's metal-rich counterpart? **[7]**

2. (a) By assuming the total pressure in a particular star is entirely due to radiation, derive the *Eddington* luminosity L_{Edd} from the equation of radiative heat transport. What is the Eddington luminosity at the surface of a $1\,\text{M}_\odot$ star with hydrogen abundance 0.7, assuming the opacity is given by electron scattering? **[6]**

(b) Show that the radiative temperature gradient $\nabla_{\text{rad}} = (d\ln T/d\ln P)_{\text{rad}}$ can in general be expressed as

$$\nabla_{\text{rad}} = \frac{1}{4}\frac{L}{L_{\text{Edd}}}\frac{P}{P_{\text{rad}}}$$

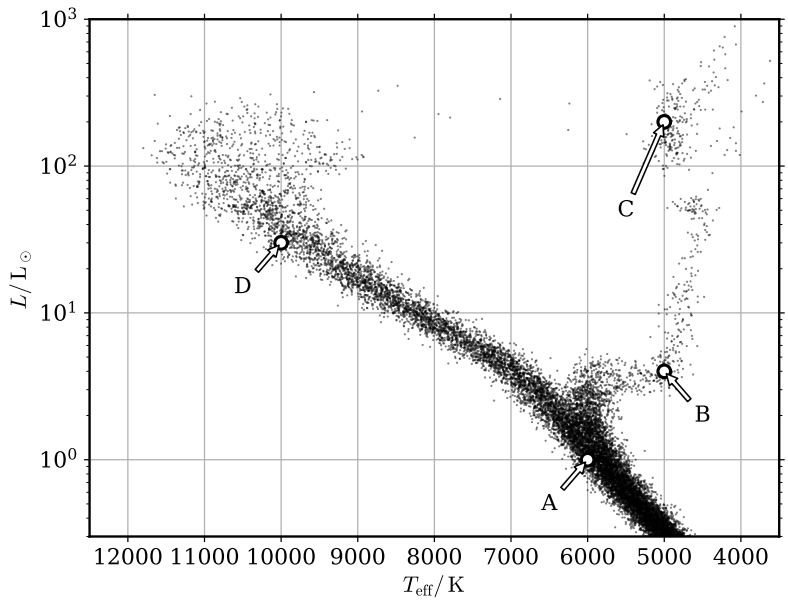
where P_{rad} is the radiation pressure. What does this imply about the convective stability of stars that are dominated by radiation pressure and have luminosities much greater than L_{Edd}? **[6]**

(c) Assuming that a small change in temperature doesn't change the entropy of radiation, what is the adiabatic temperature gradient $\nabla_{\text{ad}} = (d\ln T/d\ln P)_{\text{ad}}$ for pure radiation pressure? Derive a condition for convective stability in terms of L and L_{Edd} for a star that is dominated by radiation pressure. **[4]**

(d) Suppose that a low-mass red giant near the tip of the red giant branch has a total mass M and radius R, an envelope of constant density and a core that can be treated as a point mass with mass M_c and radius $R_c \approx 0$. Find an expression for the gravitational binding energy of the envelope. What value does this expression have just before the start of the helium flash in a $1\,\text{M}_\odot$ star with radius $176\,\text{R}_\odot$? **[10]**

(e) The first helium flash in a $1\,\text{M}_\odot$ red giant reaches a peak luminosity of $\sim 10^9\,\text{L}_\odot$ and releases about $2.5 \times 10^{48}\,\text{erg}$ of energy. Given your results above, describe why the helium flash doesn't unbind the star's envelope (i.e., everything above the hydrogen-burning shell). **[4]**

3. The figure below shows the Hertzsprung–Russell (HR) diagram of a stellar population with a roughly solar composition.



(a) Which nuclear fuels are being consumed in stars A, B, C and D? Where in each star are the reactions taking place? **[4]**

(b) Explain whether or not the stars in this population are part of a single cluster (i.e., they have a common age and composition). **[4]**

(c) Sketch the mean molecular weight μ as a function of the mass coordinate, for stars A, C and D. Assume that the initial chemical composition of each star was solar and that the stars are fully ionized throughout. Show your calculations of μ for each layer in your sketches. **[10]**

(d) Suppose that two $1.1\,\mathrm{M}_\odot$ red giants near the tip of the red giant branch merge, mix completely without losing any mass from the system, and enters a new phase of central hydrogen burning. Using the relation $L \propto M^3\mu^4$ and assuming ρ/μ is roughly constant, how much brighter and hotter is the merger product than a star of similar mass but the same Sun-like initial composition as the other stars in the cluster? Would its luminosity L and

effective temperature T_{eff} be enough to distinguish it in this population as an outlier? How else might the merger product appear different from the other stars in the cluster? **[12]**

Useful Formulae for The Life and Death of Stars

Equations of stellar structure

$$\frac{dm}{dr} = 4\pi r^2 \rho$$
$$\frac{dP}{dr} = -\frac{Gm}{r^2}\rho$$
$$\frac{dL}{dr} = 4\pi r^2 \rho \epsilon$$
$$\frac{dT}{dr} = -\frac{3\rho\kappa L}{16\pi a c r^2 T^3} \quad \text{(radiation)}$$
$$\frac{dT}{dr} = \frac{\Gamma_2 - 1}{\Gamma_2}\frac{T}{P}\frac{dP}{dr} \quad \text{(convection)}$$

Radiation constant

$$a = \frac{4\sigma}{c}$$

Mean molecular weight of a fully ionized mixture

$$\mu = \left(2X + \frac{3}{4}Y + \frac{1}{2}Z\right)^{-1}$$

Astrophysical Constants and Units

Astronomical Unit	AU	1.50×10^{11} m
Parsec	pc	3.1×10^{16} m
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_\odot$	3.84×10^{26} W
Absolute bolometric magnitude of the Sun	$M_{\rm bol}$	$+4^{\rm m}75$
Bolometric correction for the Sun	BC	$-0^{\rm m}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\rm m}74$
Solar constant	f	1.36×10^{3} W m^{-2}
Solar mass	$M_\odot$	1.989×10^{30} kg
Wien constant	b	2.898×10^{-3} m K
Hubble constant	H_o	70 km s^{-1} Mpc^{-1}
Solar radius	$R_\odot$	6.95×10^{8} m
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_\oplus$	5.972×10^{24} kg
Earth radius	$R_\oplus$	6371 km

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)